Alessia Sambruna

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Professional Objective

Graduate in statistical physics and biophysics, interested in the quantitative understanding of growth laws, gene expression, and cellular resource allocation constraints. Aims to develop mechanistic biophysical models of cell physiology, at the interface between statistical physics, stochastic modeling, and experimental data.

Education

Università degli Studi di Milano — **Italy**
Master's Degree in Physics — *Oct. 2023 – Apr. 2026*

- **Final grade**: 110/110 cum laude;
- **Thesis**: *Calibration of an Arduino-based open source automated platform to quantify Saccharomyces cerevisiae growth laws.* Calibration of the Chi.Bio system and validation of an alternative plate reader-based pipeline for the quantitative study of growth and protein expression in yeast. **Supervisor**: Prof. Marco Cosentino-Lagomarsino; **Co-supervisor**: Giorgio Tallarico (PhD student).
- **Relevant courses**: Statistical physics, biophysics, modeling of biological systems, numerical simulations.

Università degli Studi di Milano — **Italy**
Bachelor's Degree in Physics — *Oct. 2019 – Jun. 2023*

- **Final grade**: 106/110;
- **Thesis**: *Characterizing the viscoelastic properties of the mammalian cell nucleus through single molecule imaging.* **Supervisor**: Prof. Marco Cosentino-Lagomarsino; **Co-supervisor**: Prof. Davide Mazza.

Liceo Scientifico Vittorio Veneto – Milano — **Italy**
Scientific Secondary School Diploma — *2014 – 2019*
Final grade: 93/100.

Research Experience

IFOM, Milan — **Italy**
Master's Research Internship — *Mar. 2025 – Jan. 2026*
Supervisor: Prof. Marco Cosentino-Lagomarsino; **Co-supervisor**: Giorgio Tallarico (PhD student).

- Use and calibration of the Chi.Bio automated system for quantitative study of *S. cerevisiae* growth laws;
- Critical evaluation of device performance, identification of limitations for fluorescence detection (GFP);
- Development and validation of an alternative plate reader-based pipeline for joint estimation of growth rate and protein abundance;
- Quantitative analysis of relationships between cell growth, protein abundance, and nutritional conditions (Python).

Università degli Studi di Milano — **Italy**
Bachelor's Research Internship — *Nov. 2022 – May 2023*
Supervisor: Prof. Marco Cosentino-Lagomarsino; **Co-supervisor**: Prof. Davide Mazza.

- Analysis of chromosomal loci trajectories from single-molecule imaging in live cells;
- Comparison of dynamic observables with chromatin polymer models;
- Python implementation and numerical simulations.

Teaching & Outreach

Istituto Comprensivo "Paolo Neglia" Pogliano Milanese e Vanzago — **Italy**
Teaching (middle school) — *Oct. 2024 – Nov. 2024*
Teaching mathematics and science; supervision of experimental activities.

IFOM, Milan — **Italy**
Participation in the "IICII Summer School in Quantitative Biology" — *Sep. 9–20, 2024*
Attendance of seminars and practical sessions on quantitative yeast growth, including laboratory experiments and quantitative analysis.

Independent — **Italy**
Private Tutor — *2016 – Present*
Mathematics and science tutoring (20+ students).

AFS Intercultura — **Italy**
Volunteer — *2018 – Present*
Local coordinator for exchange programs (selection, training, coordination).

Italian Association of Physics Students (AISF) – Milan Committee — **Italy**
Volunteer — *2021 – 2024*
Organisation of conferences and science outreach activities.

International Experience

KU Leuven — **Belgium**
Erasmus+ Programme — *Feb. 2024 – Jul. 2024*
- **Advanced courses attended**: *Polymer Physics, Physical Modeling of Complex Systems, Advanced Statistical Mechanics, Transport Processes in Biological Systems, Stochastic Processes*;
- Participation in meetings of the *Soft Matter and Biophysics – Statistical and Computational Physics* research group.

Università degli Studi di Milano, Sorbonne Université, Heidelberg University
4EU+ Science Young To Young Curricular Project — *Oct. 2021 – Jul. 2022*
Production of a science outreach video on astrophysics and cosmology in collaboration with an international group.

AFS Exchange Program — **Ireland**
Summer Study Stays — *Jul. 2017 – Aug. 2018*
Selected through competitive process for language study stays:
- Cork (Jul.–Aug. 2018): intensive English course;
- Dublin (Jul.–Aug. 2017): participation with Intercultura scholarship, awarded by the "Associazione Culturale, Ricreativa e Sportiva dei Dipendenti del Gruppo Intesa San Paolo"; intensive English course.

Skills

Analysis & Programming: Python (regression, data analysis), image processing (ImageJ/TrackMate), simulations

Tools: Git, LaTeX, Linux, Google Colab, Microsoft Office, ECDL Certificate.

Languages: Italian (native), English (C1), French (B1), Dutch (A1+)

Research & Communication: Critical analysis, collaborative work, scientific presentation

Interests

Art and drawing: Instagram: @slowdownonpaper; **Dance**: Diploma in Vocational Graded Examination in Dance, Advanced 2 (Legat Russian Ballet Method); performer and treasurer of the theatre-dance association *Arte Liquida* since 2022; **Music**: Clarinet – member of the municipal wind band; Ukulele.